



Water dissociation at the water–rutile $\text{TiO}_2(110)$ interface from ab initio-based deep neural network simulations

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The interaction of water with TiO_2 surfaces is of crucial importance in various scientific fields and applications, from photocatalysis for hydrogen production and the photooxidation of organic pollutants to self-cleaning surfaces and bio-medical devices. In particular, the equilibrium fraction of water dissociation at the TiO_2 –water interface has a critical role in the surface chemistry of TiO_2 , but is difficult to determine both experimentally and computationally. Among TiO_2 surfaces, rutile $\text{TiO}_2(110)$ is of special interest as the most abundant surface of TiO_2 's stable rutile phase. While surface-science studies have provided detailed information on the interaction of rutile $\text{TiO}_2(110)$ with gas-phase water, much less is known about the $\text{TiO}_2(110)$ –water interface, which is more relevant to many applications. In this work, we characterize the structure of the aqueous $\text{TiO}_2(110)$ interface using nanosecond timescale molecular dynamics simulations with ab initio-based deep neural network potentials that accurately describe water/ $\text{TiO}_2(110)$ interactions over a wide range of water coverages. Simulations on $\text{TiO}_2(110)$ slab models of increasing thickness provide insight into the dynamic equilibrium between molecular and dissociated adsorbed water at the interface and allow us to obtain a reliable estimate of the equilibrium fraction of water dissociation. We find a dissociation fraction of $22 \pm 6\%$ with an associated average hydroxyl lifetime of 7.6 ± 1.8 ns. These quantities are both much larger than corresponding estimates for the aqueous anatase $\text{TiO}_2(101)$ interface, consistent with the higher water photooxidation activity that is observed for rutile relative to anatase.

water/metal oxide interfaces | deep neural network potentials | molecular dynamics simulations | TiO_2

Water is ubiquitous in the environment and its interaction with metal oxide surfaces has a key role in processes that range from wetting, dissolution, and corrosion to photocatalytic reactions (1, 2). The relative stability of molecularly vs. dissociatively adsorbed species can be of critical importance in these processes and has been intensely debated even for the simplest, low-index surfaces.

One oxide surface of major fundamental and practical interest is rutile $\text{TiO}_2(110)$ (R-110 in the following), that is widely considered the prototypical metal oxide surface (3–9). Extensive investigations of gas-phase water adsorption on R-110 have established that oxygen vacancies—the most common defects on this surface—dissociate water and become hydroxylated (10–13), while molecularly adsorbed water is only ~ 0.035 eV more stable than dissociated water at regular surface sites (14). Moreover, a $\sim 20\%$ fraction of dissociated water has been reported for one adsorbed monolayer on R-110 at $T = 210$ K (15). However, information is scarce for the interface between R-110 and liquid water, which is the system more relevant to photocatalysis and other practical applications.

Experiments based on synchrotron X-ray diffraction revealed water layering with partial dissociation of the adsorbed water on the R-110 surface (16–18). In other experiments, various ordered structures with (2×1) periodicity at the interface were reported, either on R-110 in water (19) or after exposure to liquid water (20), but such structures were found to be induced by adsorbed organic contaminants rather than interfacial water in subsequent studies (21). On the theoretical side, density functional theory (DFT)-based ab initio molecular dynamics (AIMD) simulations reported that water remains intact at the interface with defect-free R-110 (22), but this prediction could not explain the results of X-ray experiments (18).

While AIMD has provided invaluable information on the interaction of water with metal oxide surfaces (9, 23, 24), the length and time scales accessible by ab initio simulations are often insufficient to characterize the properties of aqueous oxide interfaces (25–28). For example, the duration of AIMD simulations was found to be too short to allow a reliable prediction of the equilibrium fraction of water dissociation at the aqueous anatase $\text{TiO}_2(101)$ (A-101) interface (25). To overcome these limitations, computationally efficient interatomic potentials with the accuracy of ab initio electronic structure methods

Significance

The character—molecular vs. dissociated—of adsorbed water on TiO_2 surfaces plays a critical role in many applications, e.g., photocatalysis and surface functionalization, and has been studied for decades. However, there is still limited information on water dissociation at the TiO_2 –liquid water interface, the system most relevant to applications. Ab initio simulations have provided valuable insights, but their time scales are insufficient to characterize the amount of dissociation at aqueous interfaces. We here overcome this limitation using deep neural network potentials that accurately reproduce the ab initio results for water interacting with the prototypical $\text{TiO}_2(110)$ surface. The average dissociation fraction, $22 \pm 6\%$, predicted by these simulations is a key information for understanding TiO_2 's photocatalytic activity.

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are needed. As shown by several recent studies, this goal can be achieved using ab initio-based deep neural networks (DNNs) to represent the potential energy surface of the system of interest (29–34). Still, constructing a general DNN potential for heterogeneous systems can be challenging (35), and in fact relatively few applications to aqueous metal oxide interfaces have so far been reported (27, 35–37).

In this work, we have developed and applied ab initio-based DNNs to elucidate the structure and average degree of water dissociation at the aqueous R-110 interface. Specifically, we used the “Deep Potential” (DP) scheme by Zhang et al. (30, 31) to construct a DNN potential that accurately reproduces the DFT results for both isolated gas-phase water molecules and liquid water interacting with defect-free R-110. For this surface, an additional difficulty is the significant dependence of the DFT predictions on the R-110 slab thickness (22, 38). To address this issue and be able to extrapolate the results to large slab thickness, the DP was trained on slab models of different numbers of layers and later used to run nanosecond timescale simulations of the aqueous interfaces of several different, 4 to 16 O-Ti-O trilayers thick, R-110 slabs. From these simulations, we estimated an equilibrium water dissociation fraction of $22 \pm 6\%$ and, correspondingly, a free energy difference between dissociatively and molecularly adsorbed water at the R-110/water interface of 0.040 ± 0.007 eV. Comparison of these estimates for R-110 to analogous results for aqueous A-101 (37) shows differences in the behavior of interfacial water that can significantly affect the functional properties of these two systems.

DP Training and Validation

We trained our DNN potential using potential energies and forces from periodic DFT calculations and AIMD simulations of six different systems: i) bulk rutile TiO_2 ; ii) bulk liquid water; iii) adsorbed H_2O on 4-layer R-110 $p(4 \times 2)$ and iv) 4-layer R-110 $p(6 \times 3)$ slabs; and v) the aqueous interfaces of 4-layer R-110 $p(4 \times 2)$ and vi) 6-layer R-110 $c(4 \times 2)$ slabs (*SI Appendix, Fig. S1*). All training data were generated using the Strongly Constrained and Appropriately Normed (SCAN) functional (39), which was shown to provide a good description of both TiO_2 (25) and bulk

liquid water (40) in previous studies. Details on the setup used for the DFT calculations are given in the method part and a summary of our SCAN results is given in *SI Appendix, section S1*. AIMD simulation temperatures in the range 330 to 900 K were used in order to explore a wide range of structures, including configurations with one or few interfacial water molecules either dissociated or near the transition state.

As done in previous studies of aqueous interfaces (27, 35–37), we use DPs that only consider the local environment of each atom explicitly within a finite range. While long-range contributions can in principle be incorporated in DNN potentials (41–44), they are not expected to be significant in this study since no net electric charge is present at the surface in our simulations (a condition known as point of zero charge). More importantly, numerous validation tests show that the present DPs reproduce the DFT-SCAN results with only small errors (see below and *SI Appendix*). Inclusion of long-range interactions should thus have little effect on the accuracy of the current model, which is mostly determined by the overall accuracy of the SCAN functional (39).

We assessed the quality of our DP through extensive comparison of its predicted properties for different systems to corresponding DFT-SCAN results; see, e.g., *SI Appendix, Table S5* for the lattice constants of rutile and *SI Appendix, Fig. S6* for the radial distribution functions (RDFs) of bulk liquid water. One particularly important test was the adsorption of gas-phase H_2O on the R-110 surface. It is known from previous DFT studies that the computed surface properties of R-110, including the adsorption energy of water, exhibit slowly converging even-odd oscillations with respect to the number of layers in the slab used for the calculations (22, 38). Fig. 1*A* compares DP and SCAN calculations of the adsorption energy difference ($E_d - E_m$) between dissociated (E_d) and molecular (E_m) water on slabs of different thicknesses. It appears that the oscillatory behavior of ($E_d - E_m$) predicted by SCAN [which is similar to that given by other functionals (22)] is reproduced quite well by our DP, despite the fact that only even-layered TiO_2 slabs were used for its training. While this result suggests that the even-odd oscillations of ($E_d - E_m$) are largely determined by the atomic structure of the R-110 slabs, it is remarkable that the DP is capable to capture such relationship and predict the behavior of the adsorption energies for both even- and

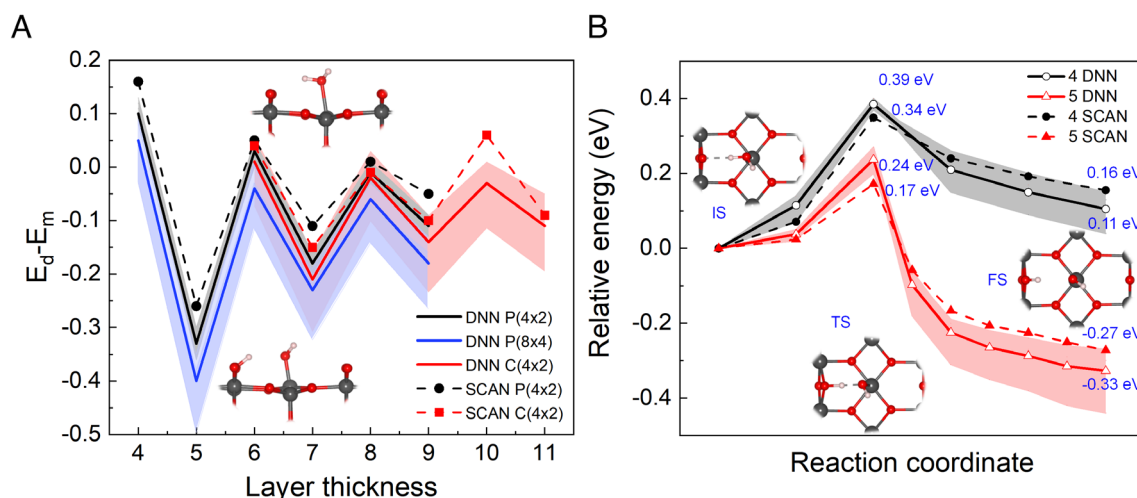


Fig. 1. Comparison between DP and DFT-SCAN results for water adsorption and dissociation on R-110: (A) adsorption energy difference ($E_d - E_m$) between dissociated and molecular water vs. number of TiO_2 layers; (B) minimum energy pathways for water dissociation on 4-layer (black) and 5-layer (red) $p(4 \times 2)$ slabs. Relevant structures are shown in the *insets*, side views in (A) and top views in (B). Molecular water is adsorbed at the surface five-coordinated Ti (Ti_{5c}) site, while dissociated water gives rise to a terminal hydroxyl at Ti_{5c} and a bridging hydroxyl at the bridging oxygen site (O_b). Dark, red, and white spheres in the model stand for Ti, O, and H, respectively. In all plots, the DP results (full lines) correspond to the average of four independent DNN potentials while the shaded areas give their SD.

odd-layered structures of increasing thickness. On a similar note, Fig. 1B compares SCAN and DP energies of selected structures along the minimum energy paths (45) to dissociate an adsorbed water molecule on R-110 slabs of 4 and 5 layers. The agreement is again quite good, even though no specific training of the DP on dissociation pathways was performed. Specifically, SCAN predicts dissociation barriers of 0.34 and 0.17 eV on the 4- and 5-layer slabs, while the corresponding DP values are 0.39 ± 0.01 and 0.24 ± 0.03 eV, respectively.

Next, we evaluated the DP's ability to describe the aqueous R-110 interface by comparing the interfacial RDFs (Fig. 2A) and water density profiles (Fig. 2B) obtained from SCAN-AIMD

and deep potential molecular dynamics (DPMD) simulations of a 4-layer R-110 $p(4 \times 2)$ slab in contact with 15 Å water. DPMD reproduces well both the water density distribution function and the RDFs at the R-110/water interface predicted by AIMD. Minor discrepancies between DPMD and AIMD are observed at surface-water distances of about 3 Å, which are however less relevant to the description of water dissociation on R-110.

Further validation tests on both gas-phase water adsorption and the aqueous interface are reported in the *SI Appendix*. Most notably, *SI Appendix*, Fig. S8 shows that the DFT result for the work to move a H^+ from a surface bridging oxygen atom to a nearby

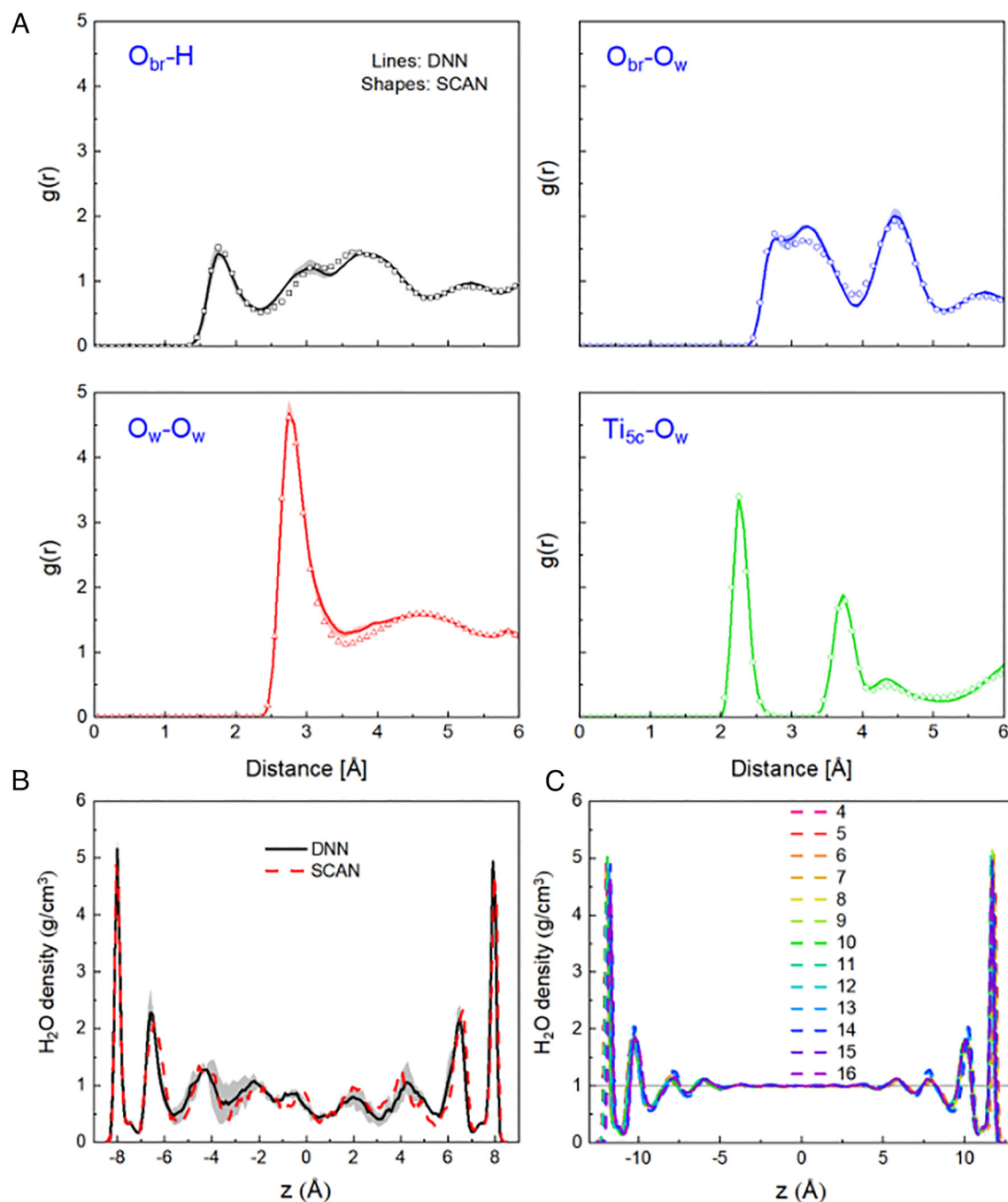


Fig. 2. Comparison between AIMD and DPMD results for (A) surface-water RDFs (surface bridging oxygen-water hydrogen, $O_{br}-H$; O_{br} -water oxygen, $O_{br}-O_w$; O_w-O_w ; $Ti_{5c}-O_w$), and (B) water density distribution of a 4-layer R-110 $p(4 \times 2)$ slab in contact with a 15 Å thick water layer. Both AIMD and DPMD simulations lasted 35 ps following 5 ps equilibration, with all water molecules initially undissociated. Shaded areas indicate the SD obtained from four independent DPMD simulations. (C) Water density distribution as a function of the number of O-Ti-O trilayers in the R-110 slab, from 20 ns DPMD trajectories following 5 ns equilibration.

terminal hydroxyl adsorbed on an adjacent Ti_{5c} atom at the aqueous interface is well reproduced by our DPs.

Finally, we compared DP- and SCAN-predicted atomic forces for all the configurations of the six different systems used to train the DP (*SI Appendix, Fig. S9*). The rmsd of the DP forces from the corresponding SCAN values vary from 70.4 meV/Å for bulk water (10,207 configurations) to 138.7 meV/Å for the $c(4 \times 2)/$ water interface (3,576 configurations). These values are quite satisfactory considering the large number of explored configurations. They are in fact in the same range of the rmsd reported by other studies employing DNN potentials (35, 46) trained on much fewer systems than the DP developed in this work.

Results and Discussion

We characterized the properties of the aqueous R-110 interface via DPMD simulations of 12 different slab models of increasing thickness, from 4 to 16 O-Ti-O trilayers, in contact with a 25 Å thick water layer between exposed surfaces. For each system, the equilibrium structure was determined from a DPMD trajectory of 25 ns duration, with the first 5 ns of simulation excluded from the statistical analysis. As shown in Fig. 2C, the water density distributions are very similar for the different slabs and overlap almost completely one with another. Similar to the structure found at the aqueous A-101 interface (37), water forms a layered structure on R-110, with the first two peaks resulting from water adsorbed at Ti_{5c} sites and H-bonded to the surface bridging oxygens, respectively. These peaks are followed by additional increasingly weaker oscillations, with the central region corresponding to bulk-like water. From the computed water diffusion coefficients (*SI Appendix, Table S8*), it appears that adsorbed water in the first layer is several times slower than bulk water, in agreement with experiment (18), while the second layer's diffusion is typically only slightly slower than in the bulk liquid and is increased by water dissociation.

The nature of interfacial water was further characterized by monitoring the spontaneous water dissociation and recombination events at each surface Ti_{5c} site during the DPMD trajectories. Such events were observed rather frequently during our 25 ns simulations (*SI Appendix, Figs. S10–S21*), and in general a larger fraction of dissociated water was found on odd-layered TiO_2 slabs than on even-layered ones, consistent with the water

adsorption energy differences in Fig. 1A. As a result, the average dissociation fractions on slabs of different thicknesses exhibit even-odd oscillations of decreasing amplitude with increasing slab thickness (Fig. 3A).

To estimate the equilibrium dissociation fraction at the aqueous R-110 interface ($f_{\text{R-110}}$), we averaged the results for the thicker slabs of 7 to 16 layers, which gave $f_{\text{R-110}} = 0.22 \pm 0.06$. This fraction can be associated to a free energy difference $\Delta A_d = -k_B T \ln(0.22 \pm 0.06) = (0.040 \pm 0.007)$ eV between dissociatively and molecularly adsorbed water at the interface, a value close to the free energy difference of 0.035 ± 0.003 eV obtained from supersonic molecular beam experiments (14) carried out at low water coverage (~ 0.05 mL) and temperature (~ 80 K). As the latter conditions are very different from those used in our simulations, the similarity of the two free energy estimates may be related to the character of the hydrogen bonds of adsorbed water at low coverage (*SI Appendix, Fig. S1*) and at the interface (Fig. 2 and *SI Appendix, Fig. S5*). In particular, the significant difference between the diffusion coefficients of first and second interfacial water layers (*SI Appendix, Table S8*) suggests that the hydrogen bonds between them are rather weak, as indeed found also for adsorbed water at low coverage.

The value of ΔA_d that we estimate for R-110/water is smaller than that recently predicted for the A-101/water interface, 0.08 ± 0.01 eV (37), consistent with the adsorption energy differences between dissociated and molecular gas-phase water at the R-110 and A-101/vacuum interfaces (12). It is known that free energy difference ΔA_d between water adsorbed in dissociated and molecular forms is related to the difference in acidities of the Ti_{5c} (pK_{a1}) and O_{br} (pK_{a2}) sites according to $\Delta A_d = 2.30 k_B T (pK_{a1} - pK_{a2})$ (47). While our simulations do not provide direct access to pK_{a1} and pK_{a2} separately, we can gain useful information by considering the point of net zero proton charge (PZC), an important quantity related to the surface acidities by $\text{PZC} = 1/2 (pK_{a1} + pK_{a2})$ (47). Following an early suggestion of slightly higher PZC of anatase than of rutile (48), a more recent extensive analysis of experimental PZC data concluded that the PZCs of rutile and anatase do not show any substantial difference, both being of the order 6 ± 1 (49). Combining this experimental result with the estimates of ΔA_d from this work and ref. 37, we then predict “optimal” pK_{a1} and pK_{a2} values of 6.35 and 5.65 for R-110, and 6.70 and 5.30 for A-101. These values are very different from available computational estimates based on short AIMD trajectories (47, 50).

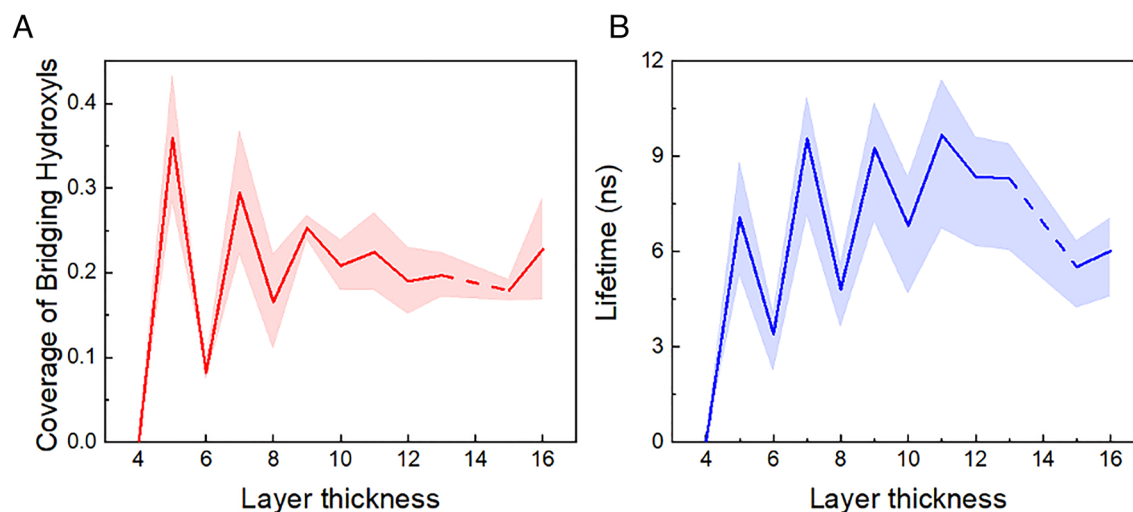


Fig. 3. Results of DPMD simulations of the aqueous interfaces of R-110 slabs with different numbers of O-Ti-O trilayers. (A) Surface coverage of bridging hydroxyls groups resulting from water dissociation; (B): lifetime of hydroxyl groups as a function of the number of O-Ti-O tri-layers. Shaded areas indicate the SD and the 95% confidence band in panels (A) and (B), respectively. All results are obtained from 20 ns DPMD trajectories after 5 ns equilibration.

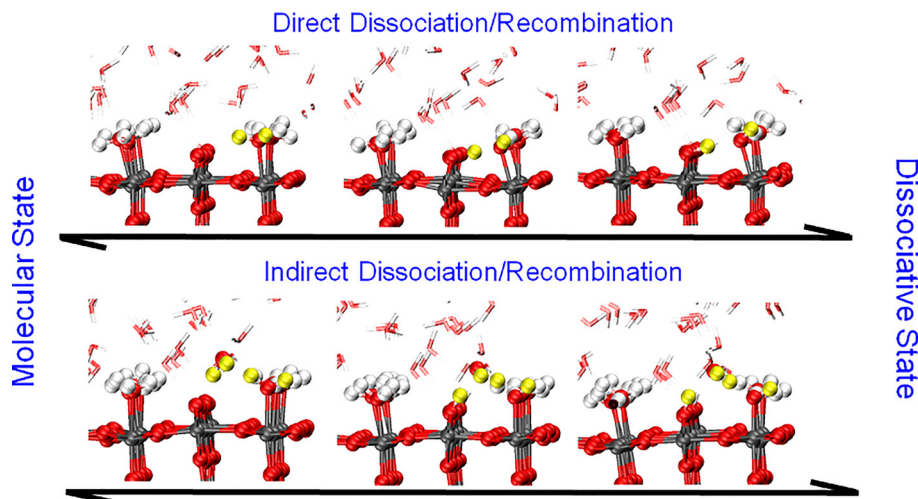


Fig. 4. Direct and indirect mechanisms of water dissociation and recombination at the aqueous R-110 interface. For better visualization, participating protons are shown in yellow, the water molecules in the first adsorbed layer are shown using balls and sticks, other water molecules are shown using sticks only. Molecular and dissociated water are on the left and right, respectively, while the middle panel is the transition state.

From the time evolutions in *SI Appendix, Figs. S10–S21* and the resulting hydroxyl survival probabilities $P(t)$ (*SI Appendix, Fig. S22*), we also calculated the lifetimes $\tau = \int_0^\infty P(t)tdt / \int_0^\infty P(t)dt$. These are shown as a function of the R-110 slab thickness in Fig. 3B, from which we estimate an average hydroxyl lifetime of 7.6 ± 1.8 ns for the thicker (7 to 16 layers) slabs. Despite the substantial error bars, both this hydroxyl lifetime and the equilibrium water dissociation fraction f_{R-110} for R-110 are much larger than the corresponding estimates for aqueous A-101 obtained in recent DPMD simulations, $\langle \tau \rangle \sim 0.6$ ns and $f_{A-101} \sim 0.06$ (37). This difference can play a significant role in photocatalysis, one of the most important applications of TiO_2 , where the transfer of photogenerated holes in TiO_2 to adsorbed water species is known to be much easier when water is dissociated (51–53). Due to the larger water dissociation fraction and longer hydroxyl lifetime, many more hole transfer processes can occur on R-110 in comparison with A-101. In particular, this can contribute to the higher water photooxidation activity of rutile with respect to anatase observed in experiment (53, 54).

Extensive studies of gas-phase water adsorption on R-110 have revealed that water dissociation occurs via direct proton transfer from the adsorbed water molecule at a Ti_{5c} site to an adjacent surface O_{br} at a distance of about 3 Å (10, 13, 14, 55–57) (Fig. 1B). An analogous direct dissociation/recombination mechanism is also observed at the aqueous R-110 interface, together with an additional indirect pathway, where a second layer water molecule donating a H-bond to an O_{br} transfers its proton to the latter and in turn receives a proton from an adsorbed water molecule at the Ti_{5c} adjacent to the O_{br} (Fig. 4). This indirect mechanism—not available when R-110 is exposed to low water coverage—is similar to that occurring at the aqueous A-101 interface (37) where the larger distance (~ 4 Å) between neighboring Ti_{5c} and O_{br} sites makes the direct proton transfer unfavorable. Interestingly, we also observed that a water molecule undergoing direct or indirect dissociation may recombine not by the same but by the other pathway, due to the fast proton diffusion at the aqueous interface.

In summary, using ab initio-based nanosecond timescale simulations we have obtained a comprehensive picture of aqueous R-110, a model photocatalytic interface (5, 13, 20), including a reliable estimate of the free energy difference between dissociatively and molecularly adsorbed water at the interface that compares well with available experimental results (14, 15)(18). In comparison with analogous results for the aqueous anatase (101)

interface (37), we found a considerably larger fraction and longer lifetime of hydroxyls on rutile (110), consistent with various known differences between the structures and chemistries of these two surfaces (9) and in particular with the higher efficiency of the rutile phase for water photooxidation (52–54). This highlights the strong link between surface structure, behavior of interfacial water, and photocatalytic activity. The present results also confirm that ab initio-based DNN potential simulations are a valuable tool for advancing the fundamental understanding of aqueous metal oxide interfaces, which is an essential step to better control the reactivity of photocatalytic and electrochemical interfaces.

Materials and Methods

DFT Calculations and Ab Initio MD Simulations. Static DFT calculations and AIMD simulations were conducted using the SCAN (39) functional, as implemented in the CP2K package (58). Goedecker–Teter–Hutter (59) norm-conserving pseudopotentials were used together with a hybrid Gaussian-plane wave basis set; for plane waves, a cutoff energy of 1,200 Ry was found necessary for convergence. The Gaussian basis sets were optimized for molecular system with double- ζ valence and one set of polarization function (DZVP) basis.

The R-110 surface was modeled using slabs with either $p(4 \times 2)$ surface supercell and 4-to-9 O-Ti-O tri-layers or $c(4 \times 2)$ surface supercell (60, 61) and 6-to-11 tri-layers (*SI Appendix, Fig. S1*). Consecutive slabs were separated by a region of at least 15 Å along the z direction. Only the Γ point was used for k -sampling. Structural optimizations were carried out until the residual forces on all atoms were less than ~ 0.02 eV/Å. The barrier of water dissociation was determined by climbing image nudged elastic band (45) calculations.

The R-110 surface in contact with liquid water was investigated via Born–Oppenheimer Molecular Dynamics simulations. The aqueous interfaces were modeled using R-110 $p(4 \times 2)$ and $c(4 \times 2)$ slabs in contact with a water layer ~ 15 Å thick, corresponding to 77 and 47 H_2O molecules, respectively, with the water density set at the experimental value of 1 g/cm^3 . The SCAN-based simulations started from structures pre-equilibrated at the PBE level (62). To sample the canonical ensemble (NVT), a Nose–Hoover thermostat (63, 64) was applied. A temperature of 330 K was used to account for the ice melting temperature of ~ 310 K predicted by SCAN (65). Hydrogen atoms were replaced by deuterium, and a timestep of 0.5 fs was used for all cases. All trajectories were run for 20 to 40 ps to ensure the interface structures are reasonably equilibrated.

DP Training and Validation. The DNN potentials were trained using the DeepMD kit (66) package, following the active learning procedure of Zhang et al. (31). The training was initiated using 100 evenly separated snapshots from the

AIMD trajectories of six different systems along with their DFT energies and forces; the six systems included: pure bulk water ($11.85 \times 12.99 \times 14.9591 \text{ \AA}^3$), bulk rutile ($3 \times 3 \times 4$ supercell), isolated water molecule adsorbed on $p(4 \times 2)$ and $p(6 \times 3)$ slabs, $p(4 \times 2)$ -R-110/water interface, and $c(4 \times 2)$ -R-110/water interface.

After training, four independent DNN potentials were generated and employed to run molecular dynamics trajectories using the LAMMPS software (67). For all systems, simulation temperatures in the range 330 to 900 K were used in order to explore wide sets of configurations. Based on the RMSD of the four DPMD trajectories, structures showing the largest deviations were selected for additional DFT calculations of energies and forces, which were then added to the training set. This procedure was repeated until the average deviation of the forces during the exploration became smaller than $0.05 \text{ eV/\AA}$ at 330 K.

The final training set included: 10,207 configurations for bulk water; 2,789 configurations for bulk rutile; 5,558 and 4,346 configurations for isolated adsorbed water on R-110 $p(4 \times 2)$ and $p(6 \times 3)$, respectively; and 11,950 and 3,576 configurations for the water interface with 4-layer $p(4 \times 2)$ and 6-layer $c(4 \times 2)$ slabs, respectively.

The four generated DNN potentials were then used for short timescale (~ 40 ps) validation studies, as described in the main text (Figs. 1 and 2). After confirming that these potentials reproduced the DFT results and were all equivalent within small error bars, one of the four DPs was selected and successively employed for all the long timescale (~ 25 ns) simulations of R-110/water interfaces.

DPMD Simulations. We considered 12 independent systems, each composed by a R-110 slab in contact with a $25 \text{ \AA}$ thick water layer at the experimental density. Each slab had a different number of O-Ti-O trilayers in the range of 4 to 16, and in all cases, we used a $p(4 \times 2)$ surface supercell exposing 8 Ti_{5c} and 8 O_{br} sites. Liquid water simulations were performed with a periodically repeated cell containing 77 water molecules at the experimental density. All simulations were performed in the NVT ensemble using LAMMPS (67). Temperature was controlled by a Nose-Hoover thermostat at 330 K (330, 400, 500, 600, 700, 800, and 900 in the exploration step). The classical equations of motion were numerically integrated using Verlet's algorithm with a time step of 0.5 fs and the hydrogen mass set to 2, which allows a larger time step without affecting the equilibrium statistical properties of the investigated systems.

Data, Materials, and Software Availability. This repository contains the training data and input files needed to train a DNN interatomic potential for TiO_2 rutile (110)/water interface. The DNN model was constructed using the DP method, as implemented in the DeepMD-kit code. Energy and forces in the data were computed with the SCAN functional implemented in the CP2K package. For more information of the methodology used to build the DNN training data, please see the reference cited at the end of this document. Within this repository you will find: raw_data: The raw data used to train the DP model; train: The DeepMD-kit input files; graphs: The frozen DNN graphs that can be used to run DP molecular dynamics. Data have been deposited in <https://github.com/CSLprinceton/Rutile-TiO-110-water> (open).

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Very recently, J. Cheng and coworkers reported DPMD simulations of the rutile $\text{TiO}_2(110)$ /water interface, also showing even-odd oscillations of water dissociation analogous to those found in our work; see: Y.B. Zhuang, R.H. Bi, and J. Cheng, *J. Chem. Phys.* **157**, 164701 (2022). However, their study predicts an equilibrium fraction of water dissociation of only 2%, which is lower than that found in our work and disagrees with several experimental evidence. This difference is possibly related to their use of a DP trained on DFT data obtained with the PBE-D3 functional, which is known to provide a description of liquid water less satisfactory than that given by the SCAN functional used in this work.

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